Ali Tarek Mohamed

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PROFILE

I'm a software engineer with a passion for building scalable and efficient full-stack applications. I specialize in both frontend and backend development, primarily using React and Express. I'm quick to learn new technologies and adapt to different frameworks and tools. My experience includes building clean, responsive user interfaces, designing normalized database schemas, and implementing secure REST APIs.

EDUCATION

Bachelor's Degree in Computer Science - Information Systems department 09/2021 – 07/2025
Cairo University

- **Relevant Courses:** Data Structures, Database Systems, Software Engineering, Operating Systems, OOP, Computer Networks, Algorithms Design and Analysis

Graduation Project (A+) 11/2024 – 06/2025

- **Problem Solving Assistant:** A cross-platform assistant for competitive programmers and CS students designed to enhance their coding skills. PSA provides personalized learning roadmaps, AI-generated solutions for coding problems, smart debugging, integration with Codeforces for progress tracking, and adaptive next-step suggestions based on user skill level.
- **Tech Stack:** React, TypeScript, Django, PostgreSQL, MongoDB, Flutter, Hugging Face, Selenium, Figma, Power BI

PROJECTS

ShopSphere (MERN Stack)

- **Live Demo**
- Built a scalable, full-stack e-commerce platform with a customer-facing storefront, admin dashboard, and RESTful API.
- Developed secure user authentication, product management, and order processing using Node.js, Express, and MongoDB.
- Designed responsive, modern UIs for both client and admin portals using React (TypeScript/JavaScript) and Tailwind CSS.
- Integrated Cloudinary for image uploads and implemented JWT-based authentication and role-based access control.

Soundora (Next.js, TypeScript, Tailwind CSS, Supabase)

- **Live Demo**
- Developed a full-stack music sharing platform where users can upload and stream audio tracks.
- Implemented Supabase authentication and storage for secure file uploads and user sessions.
- Designed a modern, mobile-friendly interface with dynamic routing and real-time track playback.
- Integrated track management with features like title, image, and genre assignment using PostgreSQL via Supabase.

Portfolio (Next.js, Tailwind CSS)

- **Live Demo**
- Built a fully responsive portfolio website showcasing projects, technical skills, educational background, and proficiency in development software.
- Styled with Tailwind CSS for a clean, maintainable design system and consistent responsive layout.

Student Management System (C++)

- Developed a streamlined student information management system using OOP principles.
- Utilized advanced data structures for efficient data handling:
  - Binary Search Tree (BST): Efficiently searched student IDs.
  - AVL Tree: Maintained balanced height for consistent operation time complexity.
  - Min Heap: Prioritized students with the lowest GPA.
  - Max Heap: Optimized access to students with the highest GPA.
- Enhanced data retrieval efficiency and accuracy, ensuring optimal system performance.

InstaChat (MERN Stack)

- Developed InstaChat, a real-time chat application
- Utilized Node.js and Express for a robust server-side architecture.
- Implemented Socket.io for real-time bidirectional communication between server and clients.
- Designed a user-friendly interface supporting private chats only, with all users as friends by default.

SKILLS

Frontend Technologies: ReactJS | NextJS | Tailwind CSS | HTML | CSS | JavaScript | TypeScript

Backend Technologies: NodeJs | ExpressJs | PostgreSQL | MongoDB

Other Tools and Knowledges: C++ | Git | GitHub | REST API | Design Patterns | SOLID Principles | Problem Solving

Familiar With: Java | Python | Docker | RabbitMq | Supabase | GraphQL | Redis | gRPC

ACHIEVEMENTS

Meta Hacker Cup Contestant 2023
- Qualified through all three rounds, achieving top 8% in the Practice Round **(948/12,138)**, top 20% in Round 1 **(3,941/20,324)**, and top 57% in Round 2 **(3,532/6,193)**.

ECPC
- Participated twice in ECPC qualifications, contributing significantly to competitive programming.

LANGUAGES

English — Fluent

Arabic — Native/Bilingual